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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert.**

I. Introduction

The rapid advancement of autonomous driving systems relies heavily on the accuracy and robustness of perception modules, particularly object detection [1], [2]. Historically, industry standards have favored closed-set object detection models, such as the widely adopted You Only Look Once architectures [3], [4]. These models are trained on large, meticulously annotated datasets to achieve real-time inference and high precision for predefined categories. However, their operational capacity is strictly bounded by the classes explicitly registered during the training phase.

Real-world driving environments are highly dynamic and unpredictable, continuously presenting edge cases and novel objects that fall outside the domain of standard training distributions. To benchmark perception models, the research community relies on various large-scale driving datasets, which can be grouped by their specific environmental focuses. For instance, Cityscapes [**?**] focuses on the high-resolution semantic understanding of complex urban street scenes across different cities. The India Driving Dataset [**?**] provides unstructured and highly unconstrained traffic environments typical of developing regions. To account for weather and lighting variations, ACDC [**?**] specifically targets adverse conditions such as fog, nighttime, rain, and snow. Meanwhile, BDD100K [5] serves as one of the most comprehensive datasets, offering diverse driving scenarios, weather conditions, and times of day to support heterogeneous multitask learning. Despite the extensive coverage of these datasets, when traditional closed-set models encounter out-of-distribution objects not present in their specific training distributions, they inherently fail to detect them or force a misclassification into a known category [6], [7]. In the context of autonomous navigation, this fundamental limitation poses severe safety hazards, as the vehicle planning systems remain blind to critical real-world obstacles and corner cases [8], [9].

To mitigate the vulnerabilities of closed-set paradigms, the field is increasingly shifting towards Vision-Language Models and Open-Vocabulary Object Detection frameworks [10], [11]. By aligning visual features with natural language text embeddings, these open-vocabulary frameworks can recognize arbitrary categories specified via text prompts without requiring domain-specific retraining. Among recent developments, YOLO-World [12] emerges as a state-of-the-art solution, combining the real-time efficiency of standard YOLO architectures with open-vocabulary, zero-shot detection capabilities. This enables the model to identify unfamiliar objects on the fly, drastically improving situational awareness.

The primary goal of this research is to comprehensively evaluate the capacity of open-vocabulary object detection systems to mitigate the inherent safety hazards posed by closed-set limitations in autonomous driving. Specifically, the zero-shot detection reliability of the YOLO-World architecture is empirically assessed against standard YOLO models when confronted with unfamiliar or unlearned objects. To conduct this evaluation, a specific data subset derived from the diverse BDD100K driving dataset, referred to as BDD10K, is utilized.utilized.

To achieve this goal, a structured experimental methodology is employed, as illustrated in Fig. **??**. Initially, the BDD10K subset undergoes necessary dataset pre-processing, including category alias mapping, to ensure label consistency, as shown in Fig. **??**a. Following this preparation, the dataset is systematically divided into known classes, such as car, bus, and truck,

Identify applicable funding agency here. If none, delete this.

used exclusively for model training, and unknown classes, including pedestrian, traffic light, and bicycle, reserved to test zero-shot capabilities. The models are then trained across multiple scales, specifically Small, Medium, and Large variants, using four distinct training configurations: two dedicated to YOLO-World training, alongside two baseline configurations for standard YOLO training, as depicted in Fig. **??**b. Finally, the performance of both YOLO-World and standard YOLO architectures is rigorously measured, as detailed in Fig. **??**c. This evaluation focuses on zero-shot metrics calculation, including mAP_{50}, alongside class-agnostic misclassification analysis. Through this methodology, the critical necessity of integrating vision-language features into autonomous perception systems to enhance real-time safety is quantitatively demonstrated.

A. Maintaining the Integrity of the Specifications

The IEEEtran class file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. You may note peculiarities. For example, the head margin measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

II. Prepare Your Paper Before Styling

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections II-A to II-H below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads—LaTeX will do that for you.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: “Wb/m^2” or “webers per square meter”, not “webers/m^2”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.
- Use a zero before decimal points: “0.25”, not “.25”. Use “cm^3”, not “cc”.)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not “Eq. (1)” or “equation (1)”, except at the beginning of a sentence: “Equation (1) is . . .”

D. LaTeX-Specific Advice

Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don’t use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in LaTeX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you’ve discovered a new method of counting.

BibTeX does not work by magic. It doesn’t get the bibliographic data from thin air but from .bib files. If you use BibTeX to produce a bibliography you must send the .bib files.

LaTeX can’t read your mind. If you assign the same label to a subsubsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

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Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won’t be any anyway) and it might stop a wanted equation number in the surrounding equation.

E. Some Common Mistakes

- The word “data” is plural, not singular.

- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word "alternately" (unless you really mean something that alternates).
- Do not use the word "essentially" to mean "approximately" or "effectively".
- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [**?**].

F. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

G. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is "Heading 5". Use "figure caption" for your Figure captions, and "table head" for your table title. Run-in heads, such as "Abstract", will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

H. Figures and Tables

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. **??**", even at the beginning of a sentence.

TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
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[a]Sample of a Table footnote.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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